

## Problem of interest

- information extraction (from pictures, audio, ... Method stay the same for each kind of data)
  - steganography: the goal is to keep all secret and information is hidden != cryptography
- mining
- recognition and classification (not only recognition, but also authentication... Is it true or not ?? - > binary classification)
- search and identification (organize data in such a way we can retrieve information... How it was made before ??? Today ??? Tomorrow ???)
- transmission and compression (smart strategy to store data)

Covariance matrix is symetric => EVD !!! For instance, a pixel is very correlated to others.

Information are concentrated in first eigen values... More correlation => eigen values decrease more fastly !!! DCT transform is very fast to compute !!! For real images, PCA tends to DCT

PCA is linear because it is only linear multiplication ! DCT vs deeplearning: easy to compute ! efficient ! deeplearning is very expansive, need some train to each dataset, time consuming etc...

- $p$  is for discrete domain. Its the probability mass function
- $f$  is for continuous domain. Its the probability density function

Be careful with llm !!! Its the most dummy tools in the world.... If not trained on something, it can make nimporte quoi !!!